



Engineering adeno-associated virus vectors for gene therapy

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Abstract | Adeno-associated virus (AAV) vector-mediated gene delivery was recently approved for the treatment of inherited blindness and spinal muscular atrophy, and long-term therapeutic effects have been achieved for other rare diseases, including haemophilia and Duchenne muscular dystrophy. However, current research indicates that the genetic modification of AAV vectors may further facilitate the success of AAV gene therapy. Vector engineering can increase AAV transduction efficiency (by optimizing the transgene cassette), vector tropism (using capsid engineering) and the ability of the capsid and transgene to avoid the host immune response (by genetically modifying these components), as well as optimize the large-scale production of AAV.

Leber congenital amaurosis
A rare genetic eye disease caused by the deficiency of various genes.

Gene therapy in its simplest form introduces genetic material into target cells, via non-viral or viral vehicles, to treat or prevent diseases by correcting or supplementing defective genes. The effects of gene therapy may be long-lasting without the need for repeated interventions, and they can be achieved via ex vivo and/or in vivo strategies. Ex vivo gene therapy uses target cells that are harvested from the patient, genetically modified and infused back into the patient. In vivo gene therapy delivers genetic material directly into the target organs or tissues of patients. Various types of gene delivery strategy have been exploited to treat a wide variety of diseases. Most diseases caused by a deficiency in a specific gene (such as haemophilia due to mutations in genes encoding clotting factors) can be corrected by delivering the missing gene. Disorders resulting from the toxicity of misfolded proteins (such as Huntington disease), can be treated by delivering nucleic acids (for example, siRNAs) that knock down the overexpressed gene (that is, RNA interference). For some disorders that are caused by unknown gene mutations, gene therapy could be used to deliver products that improve disease phenotypes (for example, anti-vascular endothelial growth factor for wet age-related macular degeneration¹, antibodies for infectious diseases² and calcium signalling proteins for congestive heart failure³).

Replacing deficient genes using gene-editing technologies, including Zinc finger nucleases, transcription activator-like effector nucleases, mega-nucleases and CRISPR–Cas9 technologies, has been explored in recent years. However, historically, therapeutic genes have been delivered to target cells as naked DNA or, more stably and efficiently owing to the ability of viruses to infect target cells, in viral vectors that have been engineered not to replicate. Indeed, in recent decades, viral vector-mediated gene therapy has been

used in clinical trials to treat cardiovascular, muscular, metabolic, neurological, haematological and ophthalmological diseases as well as infectious disorders and cancers^{4–7}. The most efficient viral vectors to emerge from preclinical and clinical studies are adenovirus vectors, adeno-associated virus (AAV) vectors, retroviral vectors and lentiviral vectors (note that lentiviruses are a subtype of retrovirus)^{6,7}. Today, AAV vectors are well established in clinical trials for in vivo gene therapy (BOX 1) and retroviral and lentiviral vectors are the vector of choice in clinical trials for ex vivo gene therapy^{4–6} (BOX 2). Furthermore, alphaviruses, flaviviruses, herpes simplex viruses, measles virus, rhabdoviruses, poxviruses, picornaviruses, Newcastle disease virus and baculovirus have all been developed as gene delivery vehicles for specific diseases or cell targeting, although their applications are limited^{5,6}.

In 2012, the first gene therapy product, Glybera, which is based on an AAV gene delivery approach, was approved in Europe for patients with lipoprotein lipase deficiency^{8,9}. In 2016, Strimvelis, the second retrovirus-mediated gene therapy product to be approved in Europe, was used to treat severe combined immune deficiency¹⁰. In 2017, the US Food and Drug Administration (FDA) approved Kymriah (tisagenlecleucel; based on a lentiviral vector) and Yescarta (axicabtagene ciloleucel; based on a retroviral vector) for the treatment of acute lymphoblastic leukaemia and large B cell lymphoma, respectively^{4,11}. Although numerous lentiviral drugs had been approved by 2017, it was not until the end of 2017 that Luxturna (voretigene neparvovec), which treats patients with RPE65-associated Leber congenital amaurosis, became the first FDA-approved AAV-based in vivo gene therapy bioproduct⁴. In May 2019, the AAV-based gene therapy drug Zolgensma (onasemnogene abeparvovec) was

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Box 1 | Results from clinical trials with AAV vectors

Following successful preclinical trials in animals, adeno-associated virus (AAV)-mediated gene delivery progressed to clinical trials with a good safety profile; promising results have been observed in patients with eye diseases, haemophilia and muscular disorders, but not in patients with cardiac diseases.

Eye diseases

Following the US Food and Drug Administration (FDA) approval of Luxturna, the first AAV-mediated gene therapy drug for patients with Leber congenital amaurosis, choroideremia, retinitis pigmentosa, Usher syndrome, Stargardt disease, Leber's hereditary optic neuropathy, wet age-related macular degeneration, achromatopsia and X-linked retinoschisis have been treated with AAV vectors in phase I/II clinical trials¹⁶⁹. AAV has been administered via subretinal and intravitreal injection; although most trials used serotypes AAV2, AAV4, AAV5 and AAV8, the mutants AAV.7m8 and AAV2tYF have also been employed^{166,169}. No severe complications related to AAV vectors manifested from these trials and a therapeutic effect was observed in some patients. However, there is variability in the visual outcome after gene therapy, and the long-term efficacy of these treatments is uncertain. In some patients, the therapeutic effect gradually decreased after reaching a peak^{169–171}; the late failure of treatment may be attributed to ongoing retinal degeneration^{170,172} or the activation of an innate immune response (as suggested by a study in primates)¹¹⁵.

Haemophilia

Phase I/II clinical trials have been carried out in the United States and Europe in patients with haemophilia A and haemophilia B using AAV2 (REF.²⁹), AAV5 (REF.³⁰), AAV6 (ClinicalTrials.gov NCT03061201), AAV8 (REFS^{27–29}), mutant AAV Spark100 (REF.²⁵) and AAV-LK03 (ClinicalTrials.gov NCT03003533) at various doses¹⁷³. Across the clinical trials with published results, an increase in clotting factor activity of 5–400% has been detected in blood. However, in some patients, elevated liver enzymes were observed 6–10 weeks after the systemic administration of AAV^{25,27–30,174}. AAV-transduced liver cells might be destroyed by a capsid-specific cytotoxic T lymphocyte response²⁸ or by activation of the innate immune response via the double-stranded RNA pathway³³.

Neurological diseases

Parkinson disease, aromatic-L-amino acid decarboxylase (AADC) deficiency, mucopolysaccharidosis (including Sanfilippo A and Sanfilippo B), Batten disease, giant axonal neuropathy, Canavan disease and metachromatic leukodystrophy have been treated with AAV vectors in clinical trials. AAV vectors including AAV2, AAV6, AAV9 and AAVrh10 (REF.¹⁷⁵) have been used to deliver therapeutic genes via intraparenchymal, intrathecal, intracerebroventricular, subpial and intravenous administration. All trials have been safe¹⁷⁵ and a therapeutic effect was observed in patients with AADC and Canavan disease using AAV2 vectors^{176,177}.

Muscular diseases

AAV gene therapy has shown efficacy in infants with spinal muscular atrophy, for which the drug Zolgensma was recently approved by the FDA³⁷. In patients with Duchenne muscular dystrophy, truncated forms of dystrophin (mini-dystrophin and micro-dystrophin) have been delivered using AAV vectors¹⁷⁸. The first trial delivered mini-dystrophin using AAV2.5 and dystrophin was detected in two patients after directed muscular injection, although an immune response to the transgene was observed²⁴. Clinical studies are underway using muscle-specific promoters with different AAV serotypes (AAV1, AAV8, AAV9 and AAVrh74) and transgenes (encoding mini-dystrophin, micro-dystrophin, β -1,4-N-acetylgalactosaminyltransferase 2 and follistatin) to treat Duchenne muscular dystrophy, and with the *MTM1* transgene (encoding myotubularin 1) to treat patients with X-linked myotubular myopathy^{178,179}.

Heart diseases

There has been limited success with the use of gene therapy to prevent and treat cardiac disorders^{3,180}. AAV vectors have been administered using intravenous delivery, retrograde delivery, antegrade intracoronary delivery, intramyocardial delivery, pericardial delivery and delivery via cardiac surgery. Although clinical trials in patients with heart failure displayed a satisfactory safety profile following the intracoronary infusion of AAV1 encoding SERCA2a, a phase IIb, multicentre, international, double-blind, placebo-controlled trial failed to demonstrate that AAV-based gene therapy was beneficial¹⁸¹.

Other disorders

The muscular delivery of AAV1 vectors encoding α 1-antitrypsin (AAT) to treat patients with AAT deficiency resulted in the expression of 2.5–3.8% of the therapeutic level of AAT required and partially corrected disease-associated defects in neutrophils¹⁸². The intravenous injection of AAV5 vectors encoding porphobilinogen deaminase (PBGD) to patients with acute intermittent porphyria, which is caused by the deficiency of this protein, reduced the number of hospital admissions and days of haem treatment needed, although the levels of the metabolic products of PBGD (namely aminolaevulinic acid and porphobilinogen) were unchanged¹⁸³.

Choroideremia

A genetic disorder with progressive vision loss due to a deficiency in Rab escort protein-1 (REP-1) owing to mutations in the *CHM* gene.

Retinitis pigmentosa

A genetic disorder that causes progressive vision loss due to inherited retinal degeneration.

Leber's hereditary optic neuropathy

An inherited mitochondrial disorder involving the loss of central vision caused by the degeneration of retinal ganglion cells and their axons owing to point mutations in mitochondrial DNA.

Achromatopsia

Autosomal recessive congenital vision loss due to malfunction of the retinal phototransduction pathway.

X-linked retinoschisis

A congenital eye disorder caused by mutations in the gene encoding retinoschisin, which plays a role in intercellular adhesion.

approved by the FDA to treat spinal muscular atrophy¹². Furthermore, the FDA released a report in 2019 predicting that 10–20 new cell and gene therapy products per year will be approved by 2025. This pace is unprecedented for new drug approvals given that small-molecule therapeutics typically take 15 years to reach approval. In addition, it highlights the advantage of drug development programmes for gene therapy, in which the missing or altered gene is known, over programmes for discovering and developing small-molecule therapeutics.

The AAV vector, which is the focus of this Review, has unique features that are beneficial in clinical applications, including broad tropism, low immunogenicity and ease of production; it is also non-pathogenic, rarely integrates into the host chromosome and results in long-term expression of the transgene^{13,14}. However, although AAV vectors have achieved early therapeutic effects in the clinic, concerns over their transduction efficiency and their tendency to induce an immune response against AAV transduced cells have been raised.

In this Review, we discuss how AAV vectors can be genetically engineered to enhance their transduction and production and to overcome immunity barriers in patients. Additionally, we propose how the efficiency of AAV vector-mediated gene therapy could be predicted in future clinical trials.

Adeno-associated virus as a vector

AAV is a single-stranded DNA parvovirus, the genome of which comprises the *rep* gene and the *cap* gene flanked by two inverted terminal repeats (ITRs)^{13,14}. The *rep* gene encodes, from a single ORF, Rep78, Rep68, Rep52 and Rep40, which aid AAV genome replication and virion assembly. Three capsid proteins (virion protein 1 (VP1), VP2 and VP3) are generated from a single *cap* ORF but are regulated by transcription from a rare start codon (ACG) and alternative splicing. As a result, VP1 and VP2 have the same amino acids as VP3 in their C-terminus. Additionally, assembly-activating protein (AAP), which is essential for capsid assembly, is encoded from an in-frameshifted ORF within the *cap* gene. All AAV virions are composed of 60 VP subunits at a 1:1:10 ratio of VP1:VP2:VP3. Each subunit has nine variable

regions on the virion surface that determine the primary tropism and intracellular trafficking of the AAV vector and are typically the domains recognized by neutralizing antibodies (NAbs)^{15,16}. Genetically modifying these variable regions can change the transduction efficiency of AAV and the ability of NAbs to bind to the virion surface^{5,7,17,18}.

In vitro studies support that AAV infects target cells by binding primary receptors and co-receptors on the cell surface, which triggers their endocytosis into endosomes^{13,19}. After a structural change exposes the N-terminus of VP1 and VP2, AAV virions are released from endosomes and accumulate at the perinuclear region of the cell^{13,20,21}. Once in the nucleus, AAV virions uncoat and release their single-stranded genome, which is converted into a double-stranded DNA (dsDNA) template from which the transgene can be transcribed and translated^{13,14} (FIG. 1).

Importantly, only the 145 bp AAV ITRs, which induce transgene expression²² and play essential roles in vector production and ensuring persistent cell transduction, are necessary for recombinant AAV (rAAV) propagation. Thus, essentially 96% of the AAV genome can be removed to permit engineering of the AAV vector for gene therapy. Indeed, substituting the *rep* and *cap* genes with an expression cassette containing a promoter (for example, the liver-specific promoter transthyretin (*TTR*)), a therapeutic transgene (for example, *F9*, which encodes coagulation factor IX (FIX)) and a poly(A) tail forms the essence of all AAV vectors²³. To date, at least 12 natural serotypes and over 100 variants of AAV have been isolated and studied as gene delivery vehicles and, from these vectors, AAV mutants have continuously been generated to optimize the use of AAV for gene delivery¹⁸. Different AAV serotypes have different binding receptors and tissue tropisms (TABLE 1), and several AAV serotypes have been used in clinical trials in patients with various diseases, including AAV1, AAV2, AAV5, AAV6, AAV8, AAV9 and AAVrh10 (AAVrh10 was isolated from rhesus (rh) monkeys), and some mutants, including AAV2.5, AAV Spark100, AAV7m8 and AAVtYF^{24–30}.

Phase I and phase II clinical trials have been carried out in patients with haemophilia B using rAAV vectors, including rAAV2, rAAV5, rAAV8 and mutant rAAV Spark100 capsids, to deliver FIX^{25,27,28}. In all studies, rAAV-mediated FIX expression was lower in patients than in preclinical animal models, even though the vector dosage per kilogram of body weight was similar^{28,31}. For example, administering 1×10^{11} particles per kg body weight of rAAV8–FIX increased FIX levels in the blood of *F9* knockout mice to 160% of FIX levels in wild-type mice. However, administering 2×10^{11} particles per kg body weight of rAAV8–FIX only increased FIX levels in the blood of primates and humans to 40% or <1% of FIX levels in healthy individuals, respectively. Furthermore, FIX levels decreased in patients treated with rAAV8–FIX after peaking but then increased following the administration of steroids^{27,28}. These studies suggest that an AAV capsid-specific cytotoxic T lymphocyte (CTL) response eradicates rAAV-transduced hepatocytes and, thus, decreases transgene expression and causes therapy to fail. Studies in mice and humans

Box 2 | Results from clinical trials with non-AAV viral vectors

Viral vectors other than adeno-associated virus (AAV), including adenoviral vectors and retroviral vectors, have been used in clinical trials.

Adenoviral vectors

Adenoviral vector, which induces transient transgene expression and is easily produced on a large scale, has a relatively large packaging capacity (8 kb), broad tropism and the ability to transduce dividing and non-dividing cells. Furthermore, infection of adenoviral vector activates innate immune signalling pathways, stimulating immune cells to secrete pro-inflammatory cytokines; these cytokines stimulate other immune cells to trigger a robust adaptive immune response. These properties make adenoviral vector a promising vaccine vehicle.

Adenovirus selectively infects and replicates in cancer cells, in which the natural expression of viral antigens or oncogenes induces the expression of pro-inflammatory cytokines that activate immune cells to kill tumour cells. As such, adenoviral vectors have been modified as oncolytic viruses that can cause the specific lysis of tumour cells and stimulate the immune system, or deliver transgenes that induce the apoptosis of cancer cells. Adenoviral vector has been applied in vaccine and cancer gene therapy in numerous clinical trials^{6,184}. The first adenoviral vector-based drug, Gendicine, was approved a decade ago and has been used in patients with cancer harbouring mutant p53. Results from clinical studies showed that Gendicine has a good safety profile and improved responses when combined with chemotherapy and radiotherapy¹⁸⁵. Recently, high-capacity adenoviral vectors (also known as gutless adenoviral vectors) were designed by deleting all viral coding sequences apart from the inverted terminal repeats and a package signal¹⁸⁴. High-capacity adenoviral vectors, which have been used to treat glioma in clinical trials (ClinicalTrials.gov NCT01811992)¹⁸⁶, can package a larger cassette (~36 kb) and show lower cytotoxicity (including a reduced immune response) than adenoviral vectors.

Retroviral vectors

Retroviruses are enveloped single-stranded RNA viruses, and retroviral vectors derived from gammaretroviruses (which infect dividing cells) or lentiviruses (which can infect dividing and non-dividing cells) have been optimized over the past three decades. The most popular retroviral vectors are based on Moloney murine leukaemia virus and HIV. Retroviral vectors integrate their DNA via reverse transcription into the host genome and have a broad tropism and low immunogenicity, meaning that they can achieve long-term expression of the target gene⁶. Retroviral vectors are mainly used clinically for ex vivo gene delivery in stem cells and chimeric antigen receptor T cell therapy, and retroviral vector-based drugs on the market include Strimvelis¹⁸⁷ (to treat severe combined immune deficiency), Kymriah¹⁸⁸ (to treat acute lymphoblastic leukaemia) and Yescarta¹⁸⁹ (to treat large B cell lymphoma).

Innate immune response

A general or non-specific defence mechanism that is the first-line defence against infection from viruses, bacteria, parasites and other foreign particles.

indicate that the level of the CTL response is dependent on the level of capsid-specific antigen presented on target cells^{27,28,32}. An innate immune response can also occur in response to double-stranded RNA (dsRNA), which can form after successful AAV transduction³³. The low transduction of rAAV in patients and the immune responses that are triggered by transduction emphasize the need to augment AAV transduction without increasing the burden of the vector capsid. Any enhancement of rAAV vector transduction will also reduce the amount of vector required for clinical trials. AAV transduction can be increased by genetically modifying the rAAV vector (by optimizing the transgene or engineering the AAV capsid)^{3,5,7,17,34} or its biological strategies (for example, by optimizing its binding to, and intracellular

trafficking and second-strand synthesis in, target cells), as discussed below.

Engineering outside the immune response

Engineering the AAV cassette

The expression of the rAAV vector upon transduction could be increased by modifying AAV ITRs so that the transgene is expressed without the need for second-strand DNA synthesis, engineering the promoter to increase transcription, optimizing the codons in the transgene to augment mRNA production and translation and improving the delivery of large transgenes (FIG. 2).

Modifying AAV ITRs. The rate of AAV transduction is limited by the fact that it requires synthesis of dsDNA from the single-stranded AAV genome before mRNA transcription can be initiated. This obligatory molecular step is initiated by the ITRs. The need for rAAV second-strand synthesis after infection can be overcome by mutating one of the wild-type ITRs. As a result, the mutated ITR is not a suitable substrate for the Rep68 and Rep78 proteins, which prevents the terminal resolution of replication and leads to the generation of specific self-complementary AAV (scAAV) replication intermediates³⁵. scAAV intermediates contain the plus and minus strands of DNA tethered by the altered ITR when encapsidated into the virion shell, unlike wild-type AAV, which packages only a single plus-strand or minus-strand DNA genome. When the two complementary halves of the scAAV cassette that are fused by the altered ITR are delivered to the nucleus, these halves anneal instantaneously to form dsDNA, which can be transcribed immediately. Of note, GFP or FIX encoded by a scAAV vector are expressed sooner, and to a higher level, than when they are encoded by conventional single-stranded AAV vectors^{23,36}. In 2020, it will be 10 years since the success of scAAV8 in clinical trials in patients with haemophilia B^{27,28}, and scAAV vectors are an integral component of Zolgensma, which was recently approved by the FDA to treat spinal muscular atrophy³⁷. Although engineering the ITR increases AAV transduction, the use of scAAV vectors is limited by the fact that they can only accommodate transgene cassettes <2.5 kb in size. Efforts to increase the packaging capacity of AAV could further enhance scAAV technology (as discussed below).

Optimizing the promoter in AAV. As the packaging capacity of AAV is limited, all *cis*-elements used, including the promoter, must be optimized. For example, long promoter sequences are substituted for small *cis*-elements that drive gene expression for the delivery of large therapeutic transgenes (such as the genes encoding coagulation factor VIII (FVIII) and cystic fibrosis transmembrane conductance regulator (CFTR)), which are 4.4–4.5 kb. Recently, different liver-specific promoter and enhancer elements were combined to generate small promoters and enhancers for the efficient packaging and transduction of rAAV³⁸. Of 42 variants tested, several were effectively packaged into AAV virions and induced high levels of expression of the transgene FVIII³⁸. In another study, screening a group of 100-bp synthetic enhancer elements

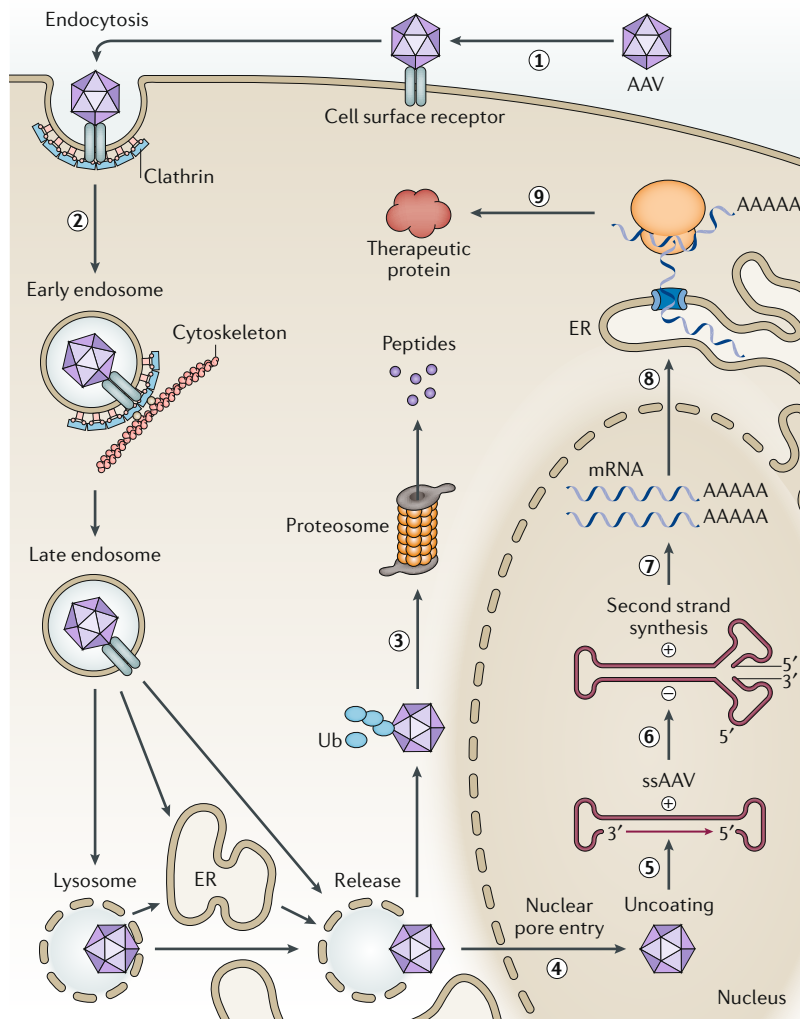


Fig. 1 | AAV vector transduction pathway. Adeno-associated virus (AAV) vector virions bind to receptors and co-receptors on the surface of target cells (step 1) and are taken into endosomes within these cells through endocytosis (step 2). Following their release from endosomes, AAV virions are either ubiquitinated and targeted for proteasome-mediated degradation (step 3) or intracellularly trafficked to the nucleus (step 4). Once in the nucleus, AAV virions are uncoated and the AAV genome is released (step 5). The AAV single-stranded DNA genome then is converted into double-stranded DNA (step 6), followed by transcription (step 7) and the nuclear export of mRNA (step 8) for translation and expression of the therapeutic transgene (step 9). Engineering AAV vectors to affect any step of their transduction pathway impacts their transduction efficiency. ER, endoplasmic reticulum; ssAAV, single-stranded AAV; Ub, ubiquitin.

Table 1 | Properties and clinical use of AAV serotypes

AAV serotype	Origin of isolation	Primary receptor	Co-receptor	Tissue tropism	Condition (ClinicalTrials.gov identifier)	Approved drug
AAV1	Monkey	Sialic acid	AAVR	Muscle, CNS, heart	Muscle diseases (NCT01519349) Heart failure (NCT01643330) AAT deficiency (NCT01054339, NCT00430768)	None
AAV2	Human	Heparin	Integrin, FGFR, HGFR, LamR, AAVR	Liver, CNS, muscle	Eye diseases (NCT00643747) Haemophilia (NCT00515710) CNS diseases (NCT00400634) AAT deficiency (NCT00377416)	Luxturna for Leber congenital amaurosis
AAV3	Human	Heparin	FGFR, HGFR LamR, AAVR	Muscle, stem cells	No trials underway	None
AAV4	Monkey	Sialic acid	Unknown	Eye, CNS	Eye diseases (NCT01496040)	None
AAV5	Human	Sialic acid	PDGFR, AAVR	CNS, lung, eye	Haemophilia (NCT03520712) Eye diseases (NCT02781480) AIP (NCT02082860)	None
AAV6	Human	Heparin, sialic acid	EGFR, AAVR	Muscle, CNS, heart, lung	Haemophilia (NCT03061201) CNS diseases (NCT02702115)	None
AAV7	Monkey	Unknown	Unknown	Muscle, CNS	No trials underway	None
AAV8	Monkey	Unknown	LamR, AAVR	Liver, muscle, pancreas, CNS	Eye diseases (NCT03066258) Haemophilia (NCT00979238) Muscle diseases (NCT03199469)	None
AAV9	Human	Galactose	LamR, AAVR	Every tissue	CNS diseases (NCT02122952) Muscle diseases (NCT03362502)	Zolgensma for spinal muscular atrophy
AAV10	Monkey	Unknown	Unknown	Muscle	No trials underway	None
AAV11	Monkey	Unknown	Unknown	Unknown	No trials underway	None
AAV12	Human	Unknown	Unknown	Nasal	No trials underway	None

AAT, α 1-antitrypsin; AAV, adeno-associated virus; AAVR, AAV receptor; AIP, acute intermittent porphyria; CNS, central nervous system; EGFR, epidermal growth factor receptor; FGFR, fibroblast growth factor receptor; HGFR, hepatocyte growth factor receptor; LamR, laminin receptor 1; PDGFR, platelet-derived growth factor receptor.

composed of ten 10 bp repeats identified an 83 bp synthetic promoter that enhanced the expression of *CFTR* (4.43 kb) in an AAV vector in human airway cell lines and in ferret airway cells in vivo³⁹.

Finally, a rational in silico approach was used to design promoters enabling the AAV-mediated liver-specific expression of transgenes⁴⁰. A genome-wide screen identified 14 hepatocyte-specific *cis*-acting regulatory modules, ranging from 41 to 551 bp long, which contained clusters of evolutionarily conserved transcription factor binding site motifs. rAAV cassettes containing these *cis*-acting regulatory module elements and a liver-specific promoter expressed the transgene in mice ~10-fold to 100-fold higher than rAAV cassettes containing the promoter alone⁴⁰. This in silico approach also identified cardiac-specific *cis*-acting regulatory modules⁴¹ and could potentially be used to design vectors with improved transduction in all tissues targeted for gene delivery. In fact, biotechnology companies have focused on optimizing promoters for gene therapy, suggesting that next-generation rAAV vectors may be derived from modular components that have

each been optimized for AAV expression and tested in clinical studies.

Optimizing the AAV transgene. The majority of therapeutic transgenes used in clinical trials are derived from natural gene sequences; the codons within them are not optimized for enhanced transduction. Traditionally, for codon optimization, the entire genomic cDNA of a host species is used to drive the codon usage bias of the organism (which typically reflects the individual tRNA concentrations)^{42,43}. Later studies observed that tRNA concentrations vary between tissues and cell types⁴⁴. To optimize constructs for gene therapy in specific tissues, one study examined tissue-specific and cell type-specific codon usage bias tables for codon optimization in the liver⁴⁵. In a human hepatocyte cell line and in mouse liver, rAAV vectors encoding FVIII from a codon-optimized sequence induced higher levels of FVIII expression than rAAV vectors containing the wild-type FVIII sequence⁴⁵. Codon usage is now standardly optimized when rAAV vectors are developed for clinical studies; numerous vendors optimize

Codon usage bias

A bias that results from background substitution biases and natural selection, and refers to the fact that, among species, some codons are more frequently used than other synonymous codons during translation.

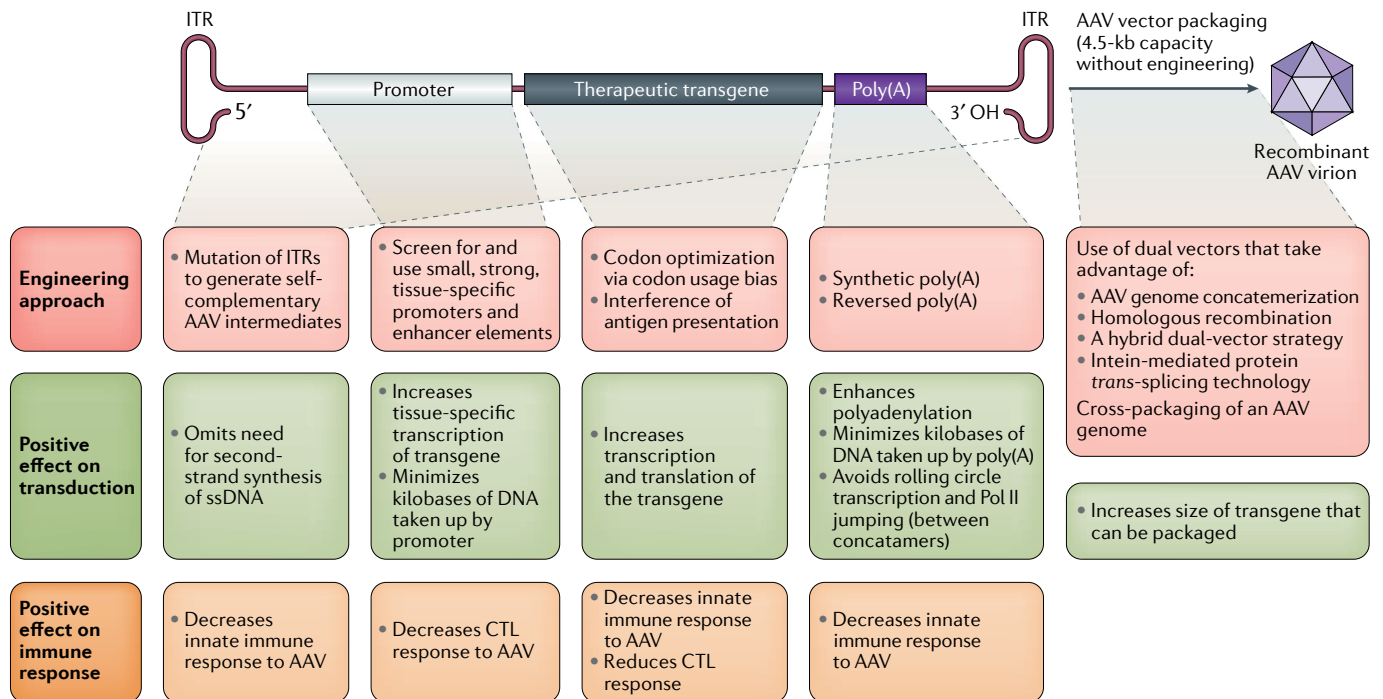


Fig. 2 | Engineering the AAV cassette. The adeno-associated virus (AAV) cassette can be engineered to enhance AAV transduction and also to enable AAV to escape immune responses. Mutation of one inverted terminal repeat (ITR) on the AAV vector, which prevents the nicking of Rep protein, can generate self-complementary AAV vector to enhance vector transduction. Mutation of ITRs may also decrease the innate response to AAV. Use of small tissue-specific promoters increases tissue-specific transgene expression and the packaging capacity of the AAV genome and minimizes the cytotoxic T lymphocyte (CTL) immune response to AAV. Optimization of transgene codons increases the transcription and translation of the AAV transgene and decreases the immune response to AAV. Using synthetic poly(A) can increase the nuclear export, translation and stability of mRNA (by enhancing polyadenylation), and using reversed poly(A) can avoid the transcription of ITR; both approaches enhance AAV transduction, and reversed poly(A) decreases the innate response to AAV. Finally, the use of dual AAV vectors or the cross-packaging of the AAV genome enables effective and functional expression of large transgenes. Pol II, DNA polymerase II; ssDNA, single-stranded DNA.

codons that can be easily expressed in AAV transgene cassettes. However, of note, a transgene optimized for maximum expression by different vendors resulted in seven cassettes for the same transgene that displayed a 2-fold to 9-fold increase in expression over wild-type sequences when tested in rAAV vectors (C.L. and R.J.S, unpublished observations). These observations suggest that our understanding of optimizing transgene codons should be viewed with caution when tested in the context of AAV vectors.

Optimizing AAV vector packaging. Many transgenes, including those encoding dystrophin (which is needed to treat Duchenne muscular dystrophy), FVIII (to treat haemophilia A) and retina-specific phospholipid-transporting ATPase ABCA4 (to treat the retinal degeneration disorder Stargardt disease), are too large to be packaged efficiently into AAV virions. Although truncated versions of several large transgenes have been used in clinical trials with some promising success^{30,46}, several other strategies can deliver larger transgenes in animals using AAV vectors.

The first approach takes advantage of the fact that, after AAV delivery, the AAV genome is concatemerized via the homologous recombination of ITR sequences^{47,48}. Large transgene cassettes can thus be split into two or

more vectors^{14,15} and delivered to the same cells; after viral uncoating in the nucleus, homologous recombination between the fragments forms an intact transgene. This approach has been used successfully in animals to deliver two or three separate AAV vectors that result in functional dystrophin^{48,49}.

In the second strategy, truncated transgene fragments are packaged into different AAV virions at undefined locations on the vector genome^{48,50}; the mixed population of AAV vectors contains truncated transgenes of different lengths. After the transduction of these dual AAV vectors, the intact transgene cassette is generated via homologous recombination of the overlapping regions of two different AAV vector genomes or by the annealing of different AAV vector genomes at complementary regions via single-stranded templates⁵⁰. Specific overlapping fragments can also be added to the end of each individual AAV vector to promote homologous recombination.

The third approach has been designed to overcome the limitations of the first two approaches; concatemerization can result in non-functional recombinant products, and the use of truncated overlapping genetic fragments can result in an unwanted ITR structure in the middle of a transgene. The hybrid dual-vector strategy combines an overlapping region and intron splice sites

Stargardt disease
A common inherited retinal disease due to mutations in ABCA4, the gene encoding ATP-binding cassette transporter (ABCA4).

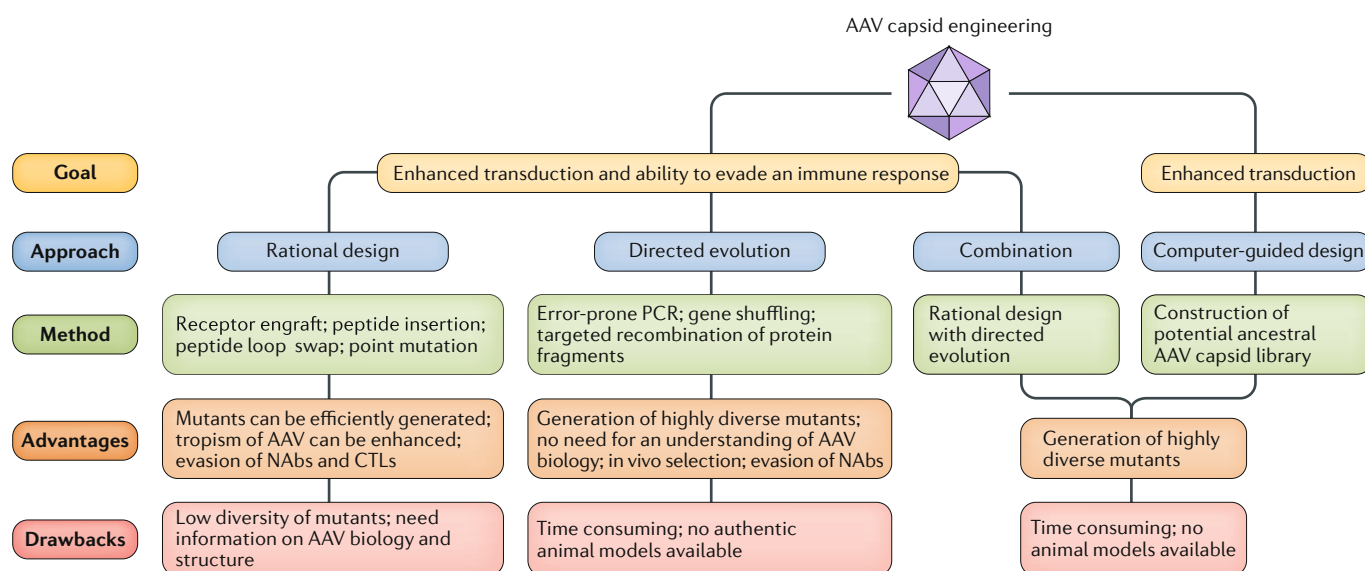


Fig. 3 | Engineering the AAV capsid. Adeno-associated virus (AAV) capsid can be engineered to enhance AAV transduction and evade immune responses. Different strategies have been explored to genetically modify the AAV capsid, including rational design, directed evolution and computer-guided design of ancestral capsid libraries. Enhanced AAV transduction can be achieved by all three of these approaches; capsids engineered with the rational design and directed evolution can also escape immune response. CTL, cytotoxic T lymphocyte; NAb, neutralizing antibody.

in the split vector transgenes⁵¹. This approach relies on the concatemerization activity of AAV genomes to bring independent AAV vector genomes together via recombination (in this case, the starting vectors are intentionally segregated into a left half and a right half carrying 5' and 3' splicing elements, respectively), and on splicing to ensure the correct transgene protein is produced after infection. One study demonstrated that the insertion of an 872 bp highly recombinogenic alkaline phosphatase sequence into AAV vectors with intron splice sites resulted in a high level of transgene-independent homologous recombination of the alkaline phosphatase sequences after transduction of the split vectors in animals⁵¹. This strategy may increase the expression of full functional protein.

A fourth approach to packaging large transgenes is to cross-package an AAV genome into the capsids of other parvoviruses. Indeed, viruses showed efficient transduction in specific cells in vitro when AAV genomes were packaged into bocavirus parvovirus⁵² and parvovirus B19 (REF.⁵³) to make chimeric vectors. Finally, using intein-mediated protein *trans*-splicing technology, split AAV vectors can be designed to package larger AAV genomes. Similar to intron-mediated RNA splicing, intein catalyses protein splicing and causes the precise ligation of two individual polypeptides through *trans*-splicing. When using this technology to deliver a larger AAV cassette, multiple AAV vectors, each encoding one of the fragments of target proteins flanked by short split inteins, are delivered to the same cells. Protein *trans*-splicing then occurs, and a full-length protein is formed. This approach has been successfully used to deliver dystrophin, FVIII, CFTR, CRISPR–Cas9, ABCA4 and centrosomal protein of 290 kDa (CEP290) in animals^{54,55}.

Although these approaches all increase the size of genes that can be packaged in AAV vectors, dual-vector

strategies, to date, are less efficient at producing a therapeutic protein than methods using a single vector, and have had variable success in animals; they may also result in the expression of unwanted products. Going forwards, these approaches may be better suited for dividing cells and transgene cassettes that need to be expressed only temporarily from an AAV vector backbone, for example, cassettes encoding CRISPR–Cas9, which may avoid off-target effects from long-term transgene expression.

Engineering the AAV capsid

Three main technologies have been exploited to engineer the AAV capsid: rational design, directed evolution and computationally designed ancestral capsids (FIG. 3).

Rational design of AAV capsids. AAV attaches to the cell surface by binding to primary receptors and/or co-receptors (TABLE 1). The receptor-binding domains on some AAV serotypes have been identified⁵⁶, and studies of AAV capsid sequences and structure have provided detailed information on the mechanisms of AAV transduction^{46,57}. This information has enabled the rational design of AAV vectors with enhanced transduction. For example, AAV1 and AAV7 have high muscle tropism, and the alignment of capsids from different serotypes, including AAV1, AAV2 and AAV7, revealed several conserved amino acid residues that may be responsible for this⁴⁶. Indeed, AAV2.5, which was generated by engrafting five conserved residues from AAV1 into the AAV2 capsid, transduced muscles more efficiently than AAV2 (REF.⁴⁶). AAV2.5 encoding mini-dystrophin has been administered by direct muscular injection in clinical trials in patients with Duchenne muscular dystrophy²⁴, validating the utility of rationally designed AAV capsids in clinical studies. AAV9 uses glycan as its primary receptor for transduction, and engrafting the glycan-binding

Mini-dystrophin

A truncated form of dystrophin that retains its function despite deletion of ~75% of the central rod domain (19 of the 24 rods; two of the four hinges) and the distal C-terminal domain (exons 71–78).

residues from the AAV9 capsid into the AAV2 capsid (generating AAV2G9) enabled AAV2 to bind glycan, thereby increasing its transduction efficiency⁵⁷. Furthermore, the rationally designed vector AAV9.HR, a version of the AAV9 capsid containing the mutations His527Tyr and Arg533Ser, retained the ability to cross the blood–brain barrier and transduce neurons after systemic administration in neonatal mice; however, its transduction was reduced in peripheral tissues⁵⁸. Based on the relationship between AAV capsid structure and function, more AAV capsids that improve vector targeting are likely to be developed⁵⁸ and may enter the clinic.

Proteasome inhibitors can enhance AAV transduction in vitro and in vivo, suggesting that modifying AAV capsids to decrease their ubiquitylation could enable AAV virions to avoid proteasome-mediated degradation in the cytoplasm. Indeed, mutating surface-exposed tyrosine residues on the AAV2 capsid prevented its phosphorylation, subsequent ubiquitylation and proteasome-mediated degradation⁵⁹. The transduction of these modified vectors was >10-fold higher than that of unmodified AAV2 in mice liver⁵⁹. Mutating surface-exposed tyrosine residues on capsids from other serotypes also enhanced AAV transduction in numerous tissues or cells in different animal models^{60,61}. AAV serotypes with tyrosine-specific mutations are entering the clinic for clinical indications, including ocular conditions (ClinicalTrials.gov NCT02416622). It will be interesting to compare the outcomes of these studies with results obtained using Luxturna, which uses a first-generation AAV2 vector developed approximately 40 years ago and is currently the only FDA-approved drug for blindness. Modifying ubiquitylation-related lysine and serine residues on AAV capsids also enhanced transduction⁶².

Rational design can also be used to generate AAV capsids that can deliver genes to cells that are not usually permissible to AAV transduction. Here, specific cell-targeting peptides are selected from a bacteriophage display library and incorporated into the AAV capsid to make AAV mutants^{63,64}. However, as this approach may alter the conformation of peptides in the virus capsid or prevent the peptide from efficiently binding to receptors on target cells, capsid variants with targeted tropism have also been selected from random AAV display peptide libraries^{17,65}. Randomized nucleic acid sequences encoding a library of peptides, including those that bind to specific receptors on target cells, are inserted into permissible sites of the AAV capsid genome, such as residue 587 or 588 of AAV2 VP1. AAV libraries derived from these mutants are made for displaying the targeting peptides on the surface of AAV virions. After infection into target cells, AAV mutants are isolated from specific tissues or cells and characterized⁶⁶. This approach has been used to isolate AAV mutants specific for various tissues or cells with templates derived from different serotypes of AAV^{17,67–69}. No additional mutations are introduced into AAV capsids aside from the inserted peptides; hence, the approach using random peptide libraries is classified as rational design. As the use of targeting peptides becomes more popular in drug delivery, we can expect increased applications with peptide insertion

in the backbone of various AAV serotypes. The crystal structures of all commonly used AAV serotypes are now available, which enables the swapping of selected or complete regions of the AAV capsid backbone between natural or chimeric variants (for example, AAV VP3 G–H domain swapping). These approaches lend themselves to parvoviruses other than AAV and may result in AAV capsids with superior transduction, tropism and immune evasion.

Directed evolution of AAV capsids. Directed evolution techniques, including error-prone PCR, gene shuffling, the targeted recombination of protein fragments, a strategy combining rational design and directed evolution, Cre recombination-based AAV directed evolution (CREATE) and in vivo directed evolution, can improve or alter the function of bioproducts and have been used to develop AAV vectors.

Error-prone PCR was first used to introduce different point mutations into the AAV2 capsid genes; AAV2 mutants with different properties, such as the ability to enhance transduction, were selected in human embryonic kidney (HEK) 293 cells⁷⁰. An AAV9 mutant library was also generated using error-prone PCR and systemically administered to mice; several AAV9 mutants, including AAV9.45 (harbouring Asn498Tyr and Leu602Phe mutations) and AAV9.61 (harbouring a Asn498Ile mutation), transduced peripheral tissues, such as cardiac and skeletal muscles, while de-targeting the liver⁷¹.

DNA sequences, and especially those linking variable regions on capsids, are highly homologous between AAV serotypes and variants. These linking sequences form the basis for gene shuffling, which results from homologous recombination between AAV serotypes. The construction of AAV libraries via gene shuffling has been described elsewhere and numerous AAV mutants have been successfully isolated from different tissues or cells in vitro and in vivo^{5,7,17,18}. Here, we summarize how this technology can be improved to develop more effective AAV vectors for the clinic.

When using AAV directed evolution, appropriate strategies for library design and selection must be used to obtain the expected variants for effective transduction. SCHEMA is a computational algorithm that can identify protein fragments, the recombination of which does not affect 3D structure, resulting in recombinant proteins that are properly folded and functional⁷². The SCHEMA-guided recombination of protein fragments from six AAV serotypes with seven crossover positions was used to generate a diversified chimeric AAV capsid library⁷³. Using a stringent Cre-dependent selection, this SCHEMA-designed AAV capsid library was administered to GFAP–Cre 73.12 mice; the variant SCH9 was isolated and found to comprise capsid fragments from AAV2, AAV6, AAV8 and AAV9 (REF.⁷³). This SCH9 variant had a higher tropism than AAV9 in adult neural stem cells⁷³. SCH9 was shown to bind to both heparin sulfate and galactose on target cells⁷³. SCHEMA-guided design of AAV capsids works owing to the availability of the crystal structures of different AAV serotypes; however, the study of AAV mutant or variant structures is far

Usher syndrome

A genetic disease caused by a deficiency in various genes that results in partial or total hearing and vision loss.

AAV helper plasmids

Plasmids containing adeno-associated virus (AAV) *rep* and *cap* genes without inverted terminal repeats.

behind the discovery of novel AAVs, which may restrict its application.

A strategy combining rational design and directed evolution was employed to develop novel AAV vectors that target mouse liver⁷⁴. Using AAV2 as a template, and based on the alignment of 150 AAV serotypes or variants, DNA sequences encoding amino acids in the variable regions of virions were mutated to increase library diversity. Surface-exposed residues were also mutated: Tyr444 and Tyr500 were mutated to reduce the proteasome-mediated degradation of the virion, and Arg585 and Arg588 were mutated to eliminate virion–heparan sulfate proteoglycan binding and increase the opportunity that AAV2 would bind to other receptors. The AAV library was constructed using AAV2 capsid as a template in combination with the mutations Y444F, Y500F, R585A and R588A and mutated variable regions from different serotypes. After selection in mice administered with the library, one AAV2-derived vector (Li-C) containing the mutations Gly263Ala, Tyr500Phe, Thr503Pro and Lys507Thr was isolated; in murine liver, Li-C had a higher transduction efficiency than AAV2 and was comparable to AAV8, the most efficient AAV serotype for transducing murine liver⁷⁴.

Although rational design or directed evolution can generate AAV vectors with enhanced transduction, these vectors might not transduce the central nervous system owing to the blood–brain barrier and the cellular heterogeneity in the brain. Thus, AAV vectors with brain tropism were developed using CREATE⁷⁵. CREATE generated an AAV library using AAV9 capsids carrying different peptides and a Cre-invertible switch. After virus libraries were systemically injected into animals with cell type-specific Cre expression, mutants including AAV-PHP.B, AAV-PHP.eB and AAV-PHP.S were isolated from, and had thus transduced, the brain of adult mice^{75,76}. However, it is concerning that these AAV variants primarily transduced neurons given that astrocyte-specific capsids were originally selected for. In addition, further analysis determined that these highly evolved tissue-specific and species-specific AAV capsids do not work in humans, although they could help dissect the mechanism of AAV biology when evaluated in animal settings.

Finally, *in vivo* directed evolution was performed to isolate AAV variants with tropism for the outer retina. Such AAV variants were created using three different libraries, namely, an error-prone AAV2 Y444F library in which AAV2 capsids contained different point mutations and the mutation Tyr444Phe, an AAV shuffled library (from AAV serotypes AAV1, AAV2, AAV4, AAV5, AAV6, AAV8 and AAV9) and a random 7mer insertion library in which AAV2 capsids contained a randomly inserted seven amino acid sequence⁷⁷. After intravitreal injection of the libraries into mice, a novel AAV variant 7m8, in which a peptide (LeuAlaLeuGlyGluThrThrArgPro) was inserted at residue 587 of AAV2 capsid VP1, was isolated from the retina and shown to transduce the retina efficiently in mice and primates⁷⁷. AAV mutants isolated using directed evolution have expanded the pool of AAV vectors available for use in future clinical trials.

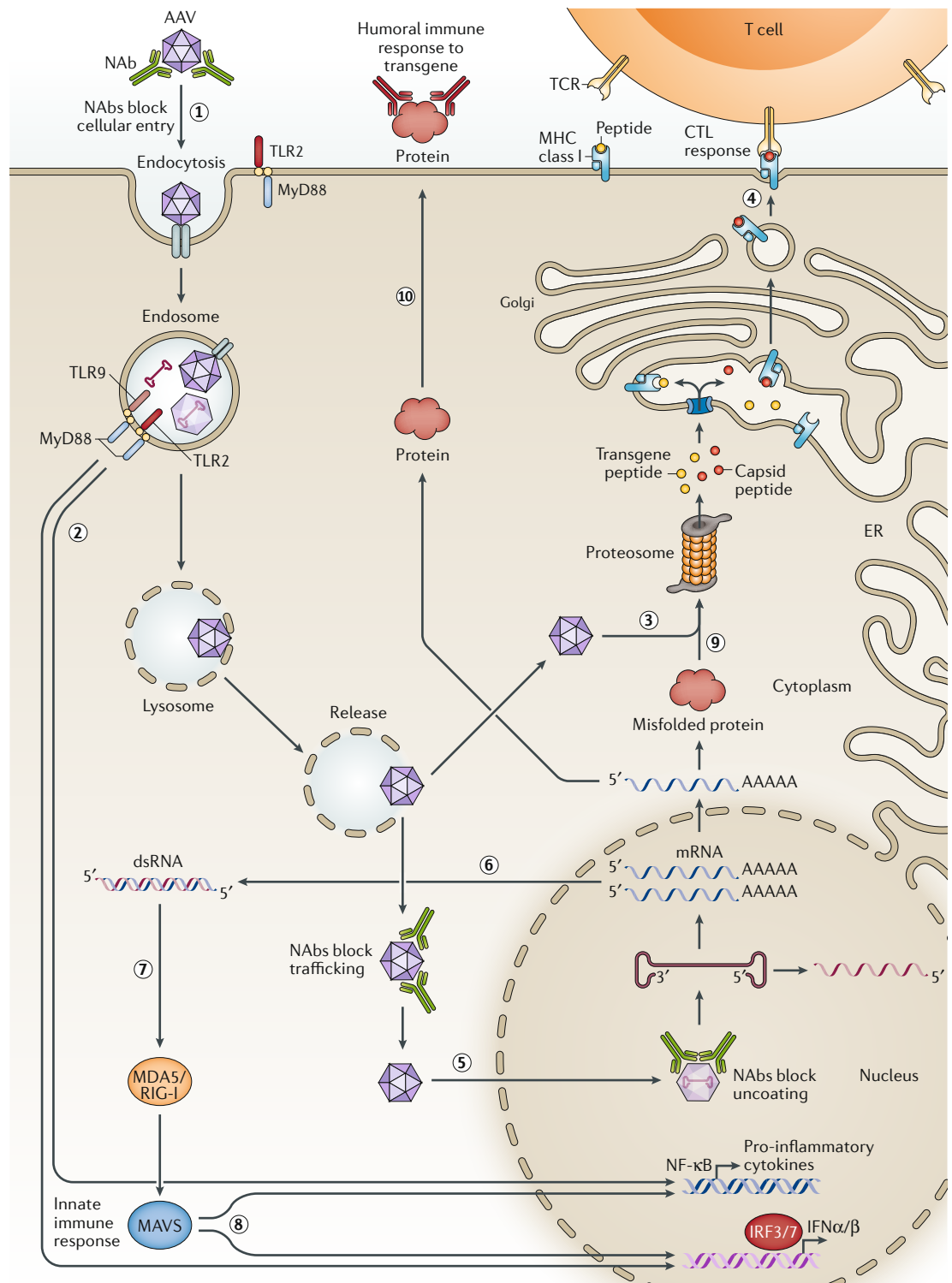
Computationally designed ancestral capsids. New capsid variants with enhanced transduction can also be generated using computational design, which uses knowledge of DNA sequences and phylogenetic analyses between AAV serotypes to construct a potential ancestral AAV capsid library^{78,79}. In total, 32 variable positions between AAV serotypes were found to be suitable for generating a library of ancestral AAV capsids⁷⁸; these capsids had similar transduction efficiencies to natural AAV serotypes in different cell lines and transduced mouse muscle more efficiently than AAV1 (REF.⁷⁸). These ancestral AAVs are also thermostable and do not use sialic acids, galactose or heparan sulfate proteoglycans as binding receptors⁷⁸, suggesting that they may transduce target cells via mechanisms different from those of unmodified AAV vectors.

Computational ancestral capsid design also identified a single ancestral AAV vector from 75 preselected AAV capsids (these capsids were preselected by integrating contemporary AAV sequence data to predict the putative ancestral amino acid sequence of AAV capsids), namely Anc80, that could transduce multiple tissues efficiently in adult mice and non-human primates, including the muscle, liver and retina⁷⁹. Anc80 could also transduce the entire cochlea and the vestibular sensory organs without any adverse effects⁸⁰, and successfully delivered *USH1C* (which encodes harmonin) to improve symptoms of hearing loss, including the partial restoration of auditory function, the startle response and balance function, in a murine model of Usher syndrome⁸¹. As additional attributes of computational AAV variants are characterized *in vivo*, it will be important to determine whether these ‘ancestral’ capsids retain critical features necessary for use in clinical studies such as the ability to be produced at high titres.

Approaches without AAV engineering

The transduction efficiency of AAV can be increased without modifying the capsid, by generating polyplex AAV vectors, using small molecules to enhance AAV trafficking or uncoating, and hijacking the function of AAV binding proteins or peptides to increase AAV transduction.

Polyplex AAV vectors. As one AAV virion is composed of 60 capsid subunits, polyplex virions containing capsid subunits from different AAV serotypes can be generated without the need for further capsid subunit modification; these polyplex vectors can have a higher transduction efficiency, greater tissue tropism and greater immune evasion ability (such as resistance to NAb) than parental serotypes. Specifically, when AAV helper plasmids (encoding AAV Rep and Cap proteins) from different AAV serotypes were mixed for transfection using a specific ratio of AAV capsid units from different serotypes^{82,83}, the transgene was more highly expressed from the polyplex vector than from parental vectors in cell lines and in mice^{46,83}, suggesting that the polyplex AAV vector has been selected for properties that enhance transduction. Current studies using combinations of structural proteins from selected serotypes (haploids) should help identify how structural



domains contribute to vector transduction (for example, VP1 and VP2 subunits from AAV8 enhance liver transduction when mixed with AAV2 VP3 subunits; C.L. and R.J.S, unpublished observations).

Small molecules. AAV transduction involves the binding of a virus to receptors followed by its release from endosomes, intracellular trafficking, uncoating in the nucleus and the synthesis of dsDNA (FIG. 1). Small molecules can interfere with these steps, especially

intracellular trafficking and second-strand DNA synthesis, to enhance AAV transduction in vitro or in animal models^{84–86}. Indeed, administration of the proteasome inhibitors bortezomib and carfilzomib enhanced the transduction of AAV in the liver of mice⁸⁵. Furthermore, a high-throughput screen of clinically used compounds identified several novel compounds that enhance AAV transduction in a vector-specific manner⁸⁷. For example, the chemotherapeutic drug teniposide enhanced the transduction of fragmented AAV2 vector compared with

◀ **Fig. 4 | Immune response to AAV vectors.** Neutralizing antibodies (NAbs) bind to the surface of adeno-associated virus (AAV) vector virions to prevent AAV virions from interacting with, and entering, target cells, and to block the effective intracellular trafficking and uncoating of AAV virions in the nucleus (step 1). For example, the antibodies A20 and ADK8 recognize and block the intracellular trafficking of intact AAV2 and AAV8, respectively. After AAV virions are taken into the cell by endocytosis, AAV capsids can be degraded in endosomes, causing the AAV genome or capsid to be exposed to receptors such as Toll-like receptor 9 (TLR9) or TLR2, triggering an innate immune response via a MyD88-driven pathway (step 2). Note that different cell types may contain different innate immune sensors. Some AAV virions do successfully escape from endosomes and enter the cytoplasm, and their capsids can be ubiquitinated and degraded in the proteasome into small peptides and epitopes (step 3). These epitopes are loaded onto the major histocompatibility complex (MHC) class I molecule and presented on the surface of the target cell, causing the cell to be recognized and, finally, eliminated by a capsid-specific cytotoxic T lymphocyte (CTL) response (step 4). For AAV virions that reach the nucleus and uncoat (step 5), transcription occurs and plus-stranded and minus-stranded RNAs are generated from both ITRs of the AAV cassette and exported into the cytoplasm to form double-stranded RNA (dsRNA) (step 6); dsRNA is recognized by the dsRNA sensors MDA5 and RIG-I (step 7), which activate an innate immune response (step 8). After expression of the transgene, any misfolded proteins are degraded in the proteasome (step 9), generating peptides and epitopes that again bind to MHC class I and trigger a CTL response against the transgene so that AAV-transduced target cells are recognized and eliminated by transgene-specific CTLs (step 4). Finally, the therapeutic product can be secreted into the circulation and taken up by antigen-presenting cells for B cell activation to produce inhibitors that neutralize the therapeutic protein via a humoral immune response (step 10). ER, endoplasmic reticulum; IFN α/β , interferon- α or β ; IRF3/7, interferon regulatory factor 3 or 7; MAVS, mitochondrial antiviral signalling; NF- κ B, nuclear factor- κ B; TCR, T-cell receptor.

its transduction without drug treatment⁸⁷. Other small molecules that enhance AAV transduction include reagents that disrupt the microtubule organization centre, an inhibitor of endoplasmic reticulum-associated protein degradation called eeyarestatin I, kinase inhibitors and the immune modulator hydroxychloroquine^{88–90}. It is expected that additional molecules that enhance AAV transduction and have similar properties to those already identified will be discovered and categorized into five groups: epipodophyllotoxins, molecules that induce DNA damage, molecules that influence epigenetic modification, anthracyclines and proteasome inhibitors. Furthermore, the efficiency of gene editing enables the genome-wide selection and refinement of key proteins involved in AAV infection and transduction (for example, trafficking proteins) that may be susceptible to the use of FDA-approved small molecules as adjuvants.

Serum proteins. In patients with disorders of the central nervous system, muscular disorders or liver diseases, AAV vectors delivered to the blood must target the whole neurological system, the entire body of muscles or the intact liver, respectively. In addition to AAV NAbs, which bind to AAV virions and block their transduction, other serum proteins may decrease^{91,92} or increase^{93–97} AAV transduction. Indeed, the direct interaction of AAV8 with the serum proteins albumin, transferrin, low-density lipoprotein and non-neutralizing immunoglobulin can enhance its transduction in the liver^{93,94,97}. Unlike other serotypes, AAV9 can cross the vascular barrier and transduce peripheral tissues after systemic injection⁹⁸. The serum proteins fibrinogen, fibronectin, von Willebrand factor, platelet factor 4, α 1-acid glycoprotein and plasminogen can interact with AAV9 to increase its

vascular permeability and transduction throughout the body⁹⁵. The direct interaction of serum proteins with AAV virions is thought to increase AAV virion–target cell binding, suggesting that AAV virions can hijack serum proteins to enhance their transduction^{93–95}.

Peptides. As AAV virions interact with serum proteins directly, they might interact with other peptides to enhance their transduction. Indeed, AAV vectors incubated with cell-permeable peptides transduced cell lines and mouse muscles (after direct muscular injection) more efficiently than AAV vectors incubated with PBS⁹⁹. Furthermore, screening several shuttle peptides for their ability to help AAV vectors across the blood–brain barrier revealed that the THR peptide interacts with AAV8 to increase its passage across the blood–brain barrier, enhancing brain transduction¹⁰⁰. Although the exact mode of action remains to be determined, these data suggest that peptides and other molecules may eventually be used in the clinic to enhance the transduction of AAV vectors. Peptides could be added to the final formulation or genetically engineered into ‘safe harbour domains’ previously identified on the AAV capsid surface. From a regulatory point of view, it may be more efficient to change the formulation (for example, by adding targeting peptides to AAV9) and proceed clinically than to characterize a new novel capsid (for example, a peptide insertion variant of AAV9).

Engineering to escape immune responses

Although AAV vectors have been successfully used in clinical studies, the host immune response is a major barrier to their ability to induce effective and long-term therapeutic gene expression. The immune response includes a CTL response to both the AAV capsid and the therapeutic protein, NAbs against AAV virions, alloantibodies (inhibitors) against therapeutic proteins and an innate immune response to AAV transduction (FIG. 4). Genetically engineering the AAV vector to evade these immune responses is of much interest.

Engineering the AAV cassette

Engineering the cassette to evade the adaptive immune response. The AAV-delivered transgene can elicit an adaptive immune response, including a CTL response and the formation of anti-therapeutic protein alloantibodies (inhibitors)^{24,101–103}. The use of a tissue-specific promoter can limit AAV transduction to target cells (for example, in the muscle or liver), preventing its expression in antigen-presenting cells and decreasing a transgene-induced CTL response¹⁰⁴. Also, as the major histocompatibility complex (MHC) class I pathway largely presents transgene products as antigens, blocking this pathway may prevent a CTL response to the transgene. Indeed, small peptides derived from viruses, such as unique short US6 glycoprotein from cytomegalovirus and infected cell protein 47 (ICP47) from herpes simplex virus, can inhibit the MHC class I pathway via various mechanisms^{105,106}. Fusing the genes encoding these viral peptides to the AAV transgene enabled cells transduced with AAV vectors to evade CTL-mediated elimination¹⁰⁷. In addition, the introduction of an

Alloantibodies

Antibodies produced from B cells after exposure to the individual's own proteins.

Major histocompatibility complex (MHC) class I pathway

MHC class I molecules are expressed on the cell surface of all nucleated cells. When peptide fragments generated from intracellular proteins bind MHC class I, the MHC class I–peptide complex is transported to the cell surface to induce the production of, and/or be recognized and killed by, cytotoxic T lymphocytes.

endogenous microRNA-mediated regulation mechanism in AAV vectors downregulated transgene expression in antigen-presenting cells and reduced the CTL response¹⁰⁸.

Targeting different AAV serotypes to the liver may induce immune tolerance in animal models¹⁰⁹. In mice, AAV8 induces stronger immune tolerance to the transgene than AAV2 after liver targeting¹⁰⁹ and less immune activation to the transgene than AAVrh32.33 after muscular delivery¹¹⁰. These findings may be due to the high level of regulatory T cells produced in blood following the targeting of AAV8 to the liver¹⁰⁹. However, on a cautionary note, recent studies demonstrated that targeting AAV8 to the liver induced the development of FVIII inhibitors in mice and dogs with haemophilia A¹⁰².

Engineering the cassette to interfere with the innate immune response to AAV. AAV virions undergo endocytosis after binding to cell surface receptors. Furthermore, Toll-like receptor 2 (TLR2) and TLR9, which are also present on the cell surface or in endosomes, can ‘sense’ AAV vectors. For example, TLR9 induced an innate immune response when dendritic cells were infected with AAV vector^{111,112}; scAAV vectors triggered a stronger response than traditional single-stranded AAV vectors¹¹². These findings suggest that the AAV capsid may possibly be degraded in endosomes, releasing the AAV genome so that it can be recognized by TLR9. Indeed, high levels of transgene expression have been observed in mice deficient for TLR9 (REF.¹¹²), prompting several groups to mutate the therapeutic cassette to decrease, or interfere with, its TLR9-mediated recognition^{113,114}. For example, the AAV transgene cassette was engineered to decrease the recognition of its CpG sites by TLR9 to avoid the activation of a TLR9-mediated innate immune response¹¹³. After muscular administration of CpG-depleted AAVrh32.33 vectors, persistent transgene expression was achieved with less infiltration from effector T cells¹¹³. This strategy has been used in clinical trials in patients with haemophilia, although not enough data are available to judge its benefits^{27,28,31}. Another strategy to interfere with the innate immune response is based on the knowledge that some oligonucleotides can interfere with TLR9–target binding. Integrating oligonucleotides, such as the (TTAGGG)₄-like sequence derived from telomeres, into the AAV cassette decreased the TLR9-mediated innate immune response and increased AAV transduction in mice¹¹⁴. Both of these engineering approaches require minimum alterations to be made to the AAV vector cassette to enhance vector transduction in animals; it is unclear whether one approach is superior to the other or whether combining these approaches would have a synergistic effect.

Interestingly, in clinical trials in patients with haemophilia, the level of transgene-encoded clotting factors decreased in some patients 6–10 weeks after the AAV vector was administered^{25,27,28}. Although it has been suggested that a capsid-specific CTL eliminates AAV-transduced hepatocytes²⁸, the role of innate immunity at the late stages of AAV transduction is poorly defined. For example, a dsRNA-mediated innate

immune response may contribute to the therapeutic failure at this late stage, owing to the fact that the AAV ITR has promoter function and the 5′ ITR and 3′ ITR transcripts can form dsRNAs; these dsRNAs are recognized by the cytoplasmic RNA sensors MDA5 and RIG-I, which are known to activate the innate immune response³³. In accordance with this hypothesis, several genes involved in the innate immune response, including *MDA5*, were upregulated in the retinal tissues of primates after long-term transduction of AAV¹¹⁵. If the ITR does help activate the innate immune response, engineering ITRs that have no or weak promoter function or engineering the therapeutic cassette to block transcription from the 3′ ITR could decrease the formation of dsRNA formation and potentially remove this concern. Alternatively, short hairpin RNAs against dsRNA sensors or their downstream signalling pathways could be cloned into the AAV cassette to block dsRNA-mediated activation of the innate immune response.

Engineering the AAV capsid

Engineering AAV capsids to evade neutralizing antibodies.

Based on published studies, >90% of humans have been infected with AAV, and ~50% of humans may have NABs^{116,117}. NABs are especially problematic for patients who require systemic administration of AAV vectors for successful treatment, such as those with neurological diseases and muscular disorders. Non-genetic approaches to reduce NAB levels include the use of pharmacological agents, such as rituximab and rapamycin, that prevent B cells from producing NABs, plasma apheresis to decrease NAB blood titres, coating the AAV surface with lipids or cell-derived extracellular vesicles to cover AAV epitopes and prevent their recognition by NABs, and the application of empty virions as decoys^{116,117}. However, as these strategies are either low in efficiency, change AAV biology or increase the AAV capsid antigen load, genetically modifying the AAV capsid to evade NABs could be beneficial, and attempts to do this using rational design and the directed evolution of AAV capsids have been made.

The AAV2 monoclonal antibody A20 recognizes multiple residues of the AAV2 capsid, including the domain around residue 265 of VP1, and blocks AAV2 transduction^{118,119}. Using rational design, the AAV2 capsid was engineered to create AAV2.5, which is mutated at residue 265 of VP1 so that it is not recognized by, or prevented from transducing cells by, A20 (REF.¹²⁰). Further knowledge of how AAV virions interact with specific monoclonal antibodies has enabled the rational design of more virions that can escape NAB activity^{121,122}. However, as antibodies in sera are polyclonal after AAV infection, not all residues recognized by AAV NABs can be identified, which limits the number of NAB-interacting residues that can be genetically modified.

Structural studies have suggested that NAB recognition sites on AAV particles might be evolutionarily conserved and located in a specific area^{77,80}. Thus, the rational mutation of residues in this region could enable the mutant virus to evade recognition by NABs. A structure-guided evolution approach, based on the specific area identified in the structural studies, evolved

Plasma apheresis

A procedure to remove the plasma from blood outside the body and reinfuse it back into patients.

murine antigenic epitopes to design AAV mutants that could evade NABs without impacting vector production, transduction efficiency or tissue tropism¹²³.

Directed evolution has also been used to isolate AAV mutants under selected pressure from NABs. For example, an error-prone PCR approach was used to generate a library with random mutations in the AAV2 capsid^{170,124}. After selection in the presence of AAV Nab-positive human serum, mutants with high tropism for HEK293 cells and the ability to escape NABs were isolated. DNA shuffling may be more efficient at isolating AAV mutants that can evade NABs than randomly mutating the AAV capsid from one serotype of AAV. An AAV shuffling library using DNA from eight wild-type viruses resulted in the isolation of AAV-DJ from primary human hepatocytes in the presence of human intravenous immunoglobulin (IVIG)¹²⁵. AAV-DJ was composed of capsids from AAV2, AAV8 and AAV9, and transduced liver more efficiently in mice treated with IVIG than the parental serotypes¹²⁵. These data suggest that it is possible to co-evolve the capsid to enhance its tissue tropism and its ability to escape NABs at the same time. Another study isolated Nab-evading AAV mutants using an AAV shuffling library in muscle in the presence of serum from patients with Duchenne muscular dystrophy after AAV treatment¹²⁶. However, the ability of these AAV mutants to transduce muscle was lower than that of AAV6, the AAV serotype most efficient at transducing mouse muscle¹²⁶. Finally, AAV mutants that can evade NABs have been isolated from humanized mouse models. Indeed, several mutants were isolated from human hepatocytes from chimeric mice carrying a human liver xenograft; these mutants escaped NABs in human sera and transduced human hepatocytes more efficiently than AAV serotypes¹²⁷.

Engineering AAV capsids to evade the CTL response. Finally, using rational design, the AAV capsid can also be engineered to avoid the capsid-specific CTL-mediated clearance of infected cells. AAV transduction results in the cross-presentation of capsid antigen on the target cell surface; this cross-presentation is mediated by the MHC class I antigen presentation pathway and requires ubiquitin-mediated degradation of the AAV capsid^{128,129}. The ubiquitylation of lysine residues on the AAV capsid can be facilitated by protein phosphorylation, and the mutation of lysine residues or of tyrosine or serine phosphorylation sites on capsids enhanced AAV transduction in cell lines and in mouse liver^{59,130,131}. Furthermore, AAV vectors in which phosphorylation sites in the capsid were mutated to avoid ubiquitylation and proteasome-mediated capsid degradation escaped the capsid-specific CTL-mediated clearance of AAV-FIX transduced target cells in mouse liver¹³¹.

All of the studies discussed in this subsection suggest that, as well as increasing transduction efficiency, capsid development will help enable AAV vectors to avoid the immune response. Whether traditional immunological drugs, AAV capsid engineering or a combination approach (immunosuppression and engineered AAV capsids) will be more effective in avoiding AAV-triggered immune responses remains to be determined.

Producing and purifying AAV vector

Optimizing AAV vector production

Producing enough AAV vector for clinical trials is challenging, especially for treating patients with neurological or muscular diseases, which usually requires $>1 \times 10^{14}$ particles per kg body weight to be administered systemically¹³². AAV vectors are commonly produced in HEK293 cells transfected with an adenovirus helper plasmid (which helps the production of AAV Rep and Cap proteins), an AAV helper plasmid (encoding AAV Rep and Cap proteins) and an AAV vector plasmid (a transgene cassette flanked by AAV ITRs)¹³³. When shaker flasks and WAVE bioreactors are used, HEK293 cells grown in suspension produce more AAV vector than adherent HEK293 cells and are most commonly used to make AAV vectors for clinical studies¹³⁴. However, the large-scale production of AAV vectors has also been tested using the stable HeLa cell line–adenovirus method, the herpesvirus helper method and the AAV Bac-based system¹³⁵. In the AAV Bac system, Sf9 cells in suspension are co-infected with three recombinant baculoviruses separately encoding the AAV Rep proteins, the AAV Cap proteins and an AAV ITR vector genome¹³⁶. Vectors generated in this system typically have less VP1 packaged into AAV virions, and a lower transduction efficiency, than those generated using other systems¹³⁷. However, VP1 production in the Bac system can be increased by engineering the sequence UGUUUUAUGUC with an attenuated Kozak sequence and leaky ribosomal scanning into the recombinant baculovirus that encodes the AAV5 cap; this engineering produces AAV vectors with a similar ratio of VP1:VP2:VP3 to wild-type vector and with a higher transduction efficiency than those made using the standard Bac system^{138,139}. Recently, the vaccinia virus was used to produce AAV vector; AAV *rep* and *cap* coding sequences were stably integrated into a single vaccinia virus, and the AAV vector cassette was cloned into a replication-deficient adenovirus from which the E1 and E3 genes had been deleted¹⁴⁰. Co-infection of these viral vectors induced the robust production of AAV virions that contained more VP1 than, and had enhanced transduction over, those produced using the traditional system, resulting in the production of $>2 \times 10^5$ AAV vector particles per cell¹⁴⁰.

As human bocavirus 1 (HBoV1) supports AAV replication it could be used to enhance AAV production^{141,142}. Indeed, co-transfecting HEK293 cells with a plasmid expressing the HBoV1 genes *NP1*, *NS2* and *BocaSR*, an AAV vector plasmid and the helper plasmid pAAVRepCap produced AAV vectors with a similar transduction efficiency to those produced using adenovirus helper instead of HBoV1 genes. Using strong promoters to increase the expression of AAV capsid proteins and Rep52 in the helper plasmid increased the yield of rAAV when co-transfected with the HBoV1 genes and AAV vector plasmid¹⁴¹. Furthermore, rAAV vector production was three to seven times higher when the adenovirus *E2A* gene was included in the HBoV1 helper system compared with when it was encoded by adenovirus helper¹⁴¹. These data suggest that the complex relationship between AAV and its natural helper adenovirus, and the role of various viruses in enhancing

Adenovirus helper plasmid

A plasmid containing most adenovirus genes that helps the production of adeno-associated virus (AAV) Rep and AAV replication.

Stable HeLa cell line–adenovirus method

A HeLa cell line contains adeno-associated virus (AAV) *rep* and *cap* genes, with or without integration of the AAV vector genome. When transduced with wild-type adenovirus and AAV vector, AAV Rep and Cap will be produced and the AAV vector genome will replicate and be packaged to produce a large amount of AAV vector.

Herpesvirus helper method

Recombinant herpesvirus vectors (one contains adeno-associated virus (AAV) *rep* and *cap*, another contains AAV vector genome) are used to deliver the *rep* and *cap* genes, as well as the AAV vector genome, into HeLa cells for AAV vector production. Helper genes for helping AAV Rep and Cap production are provided by the herpes simplex virus genome.

Kozak sequence

A sequence with the consensus ACCAUGG and a critical role in translation initiation.

Leaky ribosomal scanning

A mechanism for regulating gene expression during the initiation phase of eukaryotic translation, in which a suboptimal translational initiation codon on mRNA is skipped by the small 40S ribosome subunit in translation initiation.

the production of AAV vectors, needs to be understood further before these approaches can be used to generate AAV vectors for clinical use.

Optimizing AAV vector purification

There are currently two techniques for purifying AAV: density gradient centrifugation and chromatography purification¹³⁵. Combining these methods ensures an AAV product of high purity. Several ligands for AAV have been used for affinity chromatography, including heparin¹⁴³, mucin¹⁴⁴, A20 monoclonal antibody¹⁴⁵ and the resin AVB Sepharose¹⁴⁶; using epitopes on the AAV capsid surface to purify AAV has also been explored. For example, AAV vectors can be biotinylated or engineered to encode a hexa-histidine tag and purified with avidin or Ni-NTA columns, respectively^{147,148}. The affinity of some AAV serotypes for AVB Sepharose can be enhanced, by incorporating key residues from AAV serotypes with the highest affinity for AVB Sepharose into them, without compromising the transduction efficiency of the vector¹⁴⁹. As AAV engineering generally does not markedly change the molecular weight and structure of AAV, genetically modifying the AAV capsid is unlikely to impact density gradient centrifugation, although it could impact column chromatography affinity.

Of note, as empty virions and vector-containing virions have an identical amino acid composition, affinity chromatography cannot discriminate between them, although ion exchange chromatography can¹⁵⁰. Empty AAV capsids have been used as a decoy to prevent NABs from recognizing vector-containing particles¹⁵¹, although they increase the antigen load from AAV capsids, which may induce an unnecessary immune response¹²⁹. Recently, a novel platform with a two-column purification method that uses affinity chromatography and ion exchange chromatography was developed, and it is compatible with a range of AAV serotypes¹⁵². This approach generated AAV vectors of high purity with less contamination by empty capsids than other approaches¹⁵².

Evaluating gene therapy efficiency

Although progress has been made in the engineering of AAV vectors, and especially in the genetic modification of the AAV capsid, novel AAV mutants that show positive results in some animal models may not be effective in other species^{28,31,75,153–156}. However, as testing the transduction efficiency of AAV vectors in humans is impractical and unethical, the development of systems that can evaluate AAV transduction and predict the efficacy of vectors in human trials is needed.

Animal models

Although animal models have been used to evaluate the therapeutic efficacy, toxicities and immune response to AAV gene therapy, results from mouse models do not extrapolate to large animals, humans or even other strains of mouse^{28,31,75,153–156}. For example, a similar dose of AAV vector to that used to treat haemophilia in mice yielded a 10-fold and 100-fold lower therapeutic effect in large animals and humans, respectively^{28,31}. Furthermore, the AAV PHP.B vector isolated from the

brain of C57BL mice transduced the brain of these mice more efficiently than AAV9, but did not improve virus transduction to the brain of other mouse strains or animal species^{75,153,154}. To overcome such issues, humanized mouse models were developed to test the transduction efficiency of AAV serotypes and mutants in mice harbouring human tissue^{127,157–160}. However, early studies in these models showed that AAV8 had a lower transduction efficiency than other AAV serotypes, including AAV3 (REF.¹⁵⁷), whereas other studies showed AAV3 and AAV8 to have similar transduction efficiency in human hepatocytes in chimeric mice^{157,159}. A recent study showed that AAV7 transduced human hepatocytes more efficiently than other serotypes, including AAV3 and AAV8 (REF.¹⁶¹). Several differences between these studies might explain these discrepancies, including the time after AAV administration when AAV transduction was measured, the virus purification approach used, the difference in human hepatocyte donor, whether nitisone treatment (which prevents the death of hepatocytes in fumarylacetoacetate hydrolase-null mice) was used, the engraftment efficiency of the human hepatocytes, the number of mouse hepatocytes remaining after engraftment, and whether transduction efficiency was assessed via flow cytometry, microscopy or by gene copy number^{157–162}.

Aside from liver cells, human muscle cells and islets have been grafted into immune-deficient mice^{163,164}. However, although the ability of AAV1 to transduce mice harbouring human muscles has been investigated¹⁶³, the transduction efficiency of different AAV serotypes in human muscles and islets in xenograft mouse models has not been reported.

Finally, it is unclear whether results gained in humanized mouse models can predict the transduction efficiency of AAV in humans; mice carrying xenografts from other species can help address this question. For example, the transduction efficiency of AAV in mice engrafted with primate hepatocytes can be compared with the transduction efficiency of AAV in primates themselves. Such experiments may indicate how useful mouse models carrying human tissue grafts are in assessing AAV delivery systems.

Organoids or 3D tissue culture

Organoids or 3D tissue culture systems may be used to assess the efficiency of AAV transduction. Organoids are in vitro 3D miniature organs formed from the differentiation of specific cells¹⁶⁵ that have been established to study the pathogenesis and treatment of disease. As organoids can be generated from human cells, they may reflect the physiological conditions of humans better than animal models¹⁶⁵. Several organoids have been generated in the laboratory and transduced with AAV: AAV8 and AAV9 successfully transduced photoreceptors in organoid models^{166,167}. Furthermore, a liver organoid was used to compare the transduction of different AAV serotypes and mutants¹²⁷; the transduction efficiency of these AAV vectors differed between human liver organoids and humanized mice, highlighting the variability between systems for evaluating novel AAV capsids¹²⁷. To date, all organoid models exhibit a

Complement activation

Complement is a system made up of plasma proteins that can be activated by a pathogen or the antigen–antibody complex. Complement activation enhances the ability of antibodies or phagocytic cells to clear invading microorganisms or damaged cells.

fetal-like tissue phenotype, which may limit their use in evaluating the efficiency of AAV transduction.

Conclusions and perspectives

Although AAV gene therapy has shown promise in clinical trials, it faces challenges, including lower transduction efficiency in humans than in animal models, an immune response to AAV capsids or transgenes, inefficient methods for virus production and purification, and a lack of reliable systems to evaluate the efficiency of AAV transduction for human clinical trials. The genetic modification of AAV vectors has the potential to overcome these obstacles. Indeed, engineered AAV capsids can increase the infectivity of AAV and enable the virus to escape NABs, and modified AAV transgene cassettes can augment transgene expression and prevent the immune response to AAV capsids and transgenes. However, although AAV vectors with engineered capsids have been developed in vitro or in animal models, their advantages may not transfer to patients, as observed in the clinic for haemophilia (over seven clinical trials are still currently evaluating the best AAV-mediated approach for delivering gene therapy to patients with haemophilia) and other diseases. The inability to reproduce data in current animal models urges us to develop systems that more accurately predict the transduction efficiency of engineered AAV vectors in human target tissues.

Based on their safety profile and therapeutic efficacy, AAV vectors seem to be a promising gene delivery vehicle. Two AAV-based drugs (Luxturna and Zolgensma) have been approved by the FDA, and clinical trials using AAV-based therapies are in progress in patients with genetic and non-genetic disorders. Information gained from these trials, and from the long-term follow-up of the use of these approved drugs in the next 5–10 years, will help direct the next steps for the field. However, we are entering a vortex of vector development that is outpacing clinical evaluation. For example, the original AAV2, which is the only AAV serotype approved for ocular gene delivery, may not be considered for gene therapy trials in the eye today based on accumulated data gained in small and large animals and from donor tissues. Yet it may be 3–5 years before any of the promising engineered AAV vectors available today enter the clinic, if they ever do. The innovation of AAV vectors may soon increase further owing to the fact that the majority of natural AAV capsid vectors are coming off

patent and are likely to have FDA validation in humans; these facts could make it even more difficult for investigators to risk potential therapeutic success with human tested capsids compared with novel next-generation capsids. Although results from studies in animals suggest that there is now a superior AAV reagent for each first-generation AAV vector currently being used in clinical trials, there is no proof that these reagents will be better than AAV parental serotypes in humans; however, clinical trials using several AAV mutants are underway. The use of humanized mice models or of 3D human cultures provides a new paradigm for testing AAV capsids, but drawbacks to these assays are likely to be unveiled. It seems possible that engineering AAV vectors at the transgene level, for example, to optimize codons, promoters and *cis*-elements, may have the greatest potential to positively impact all AAV vectors in the clinic. Engineering these transgene variables affects all species at the fundamental level (microRNA, DNA replication, recombination, promoters, transcription, RNA stability and so on), all of which are more evolutionarily conserved between species than the virus capsid tropism. Improvements in large-scale AAV production are also likely to have a positive impact on the use of AAV vectors in the clinic in the near future. These improvements will dominate the AAV landscape over the next 5 years as approaches to bring down production costs are emphasized going forwards.

Results from clinical trials are raising, and will continue to raise, concerns that were not clearly demonstrated in preclinical studies or in the majority of current clinical studies (for example, high doses of AAV lead to complement activation)¹⁶⁸. The ability to modulate the immune system to improve vector delivery, to suppress unwanted immune responses or to include more patients in ongoing trials, based on the exclusion criteria for NABs, are all promising and are likely to be at the forefront of future research in the field.

In short, we have come a long way from ‘bench to bedside’ given the newly approved FDA drugs based on the AAV vector. Ultimately, AAV vectors may only be a valuable clinical tool for a brief moment in time as we look to the future of human gene therapy and envision technological improvements that come closer to the development of synthetic biological nanoparticles that mimic the many attributes of virus particles used today.

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Author contributions

Both authors contributed to all aspects of the article.

Competing interests

R.J.S. is the founder and a shareholder at Asklepios BioPharmaceutical and Bamboo Therapeutics, Inc. He holds patents that have been licensed by University of North Carolina to Asklepios Biopharmaceutical, for which he receives royalties. He has consulted for Baxter Healthcare and has received payment for speaking. C.L. is a cofounder of Bedrock Therapeutics, Inc. He holds patents licensed by University of North Carolina and has received royalties from Bedrock Therapeutics and Asklepios Biopharmaceutical.

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